



## CCNP Lab: Multi-Area OSPF



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**Purpose:**

The purpose of this lab is to learn how to install a multi-area OSPF routing topology with up to 8 routers. Furthermore, it also teaches what the purpose of multi-area OSPF is in comparison to other OSPF topologies such as single-area OSPF.

**Background:**

OSPF (Open Shortest Path First) is a popular Interior Gateway Protocol (IGP) developed by the Internet Engineering Task Force (IETF). It was first created in 1989 in RFC 1131 and later optimized and updated in RFC 2328 and RFC 5340. Previous routing protocols had slow convergence and scalability issues such as RIP (Routing Information Protocol), which inspired OSPF to be created to combat this problem. OSPF is a link-state routing protocol which means routers maintain a map/topology of the network and use Dijkstra's algorithm to compute the shortest possible route for routing.

The primary goals of OSPF are fast convergence, scalability, support for hierarchical network design, efficient use of bandwidth, and classless routing.

In Single-Area OSPF, all routers are placed in the backbone area, which is usually Area 0. This means that every single router has the same Link-State Database and has the benefit of being strong for networks with fewer routers. The issue with this topology is as the network grows, the Link State Data Base similarly grows, which causes high memory and CPU usage once the topology scales are large enough. Furthermore, a change in one router can trigger SPF recalculations across all routers.

Multi-Area OSPF was introduced to combat the scalability problems associated with single-area OSPF. It allowed the administrator to break the network into multiple different areas, with one being the backbone area (Area 0). All areas are connected to the backbone area with each area having its own Link-State Database. Only necessary routing information is shared between areas which reduces the size of the routing table.

As a result of the introduction of Multi-Area OSPF, large-scale environments saw improved scalability and convergence, alongside more logical segmentation which allowed technicians to troubleshoot and manage the network easier.

In the modern day, Multi-Area OSPF is common in data centers, WANs, and campus networks where hierarchical design is critical with optimal performance. Oftentimes Multi-Area OSPF is used alongside BGP for external routing since OSPF handles internal routing.

In this lab, we used a topology of 5 routers with two multi-layer switches. Multi-layer switches are switches on the OSI Layer 2 which generally act like an ordinary network

switch, except that they provide extra functions for higher OSI Layers, mainly Layer 3 in this case. It was invented by engineers at Digital Equipment Corporation, and in the case of layer 3 switches, its purpose is to lower network latency due to the fact that a packet can be routed without making extra network hops to a router. Oftentimes as a result of their simplifying nature, they oftentimes are used as a cheaper, lower latency replacement in many networks.

Despite their strong benefits for networks, they carry many limitations. For instance, many L3 switches either don't support NAT or have very limited capabilities, and advanced routing protocols alongside other router-based functionalities such as VPN, Tunneling, WAN Connectivity, and more are very difficult to do with a L3 switch and as such need a router instead.

### **Lab Summary:**

In this lab, I installed Multi-Area OSPF on a topology which consists of 5 routers connected in a line, with two multi-layer switches at the end, for a line of 7 machines configured with multi-area OSPF. I configured the first two routers in this line to be in Area 1, then the next three routers to be in Area 0, and finally the last two multi-layer switches to be in Area 2. I assigned unique networks for each router in the 10.0.x.x network with unique router ID's, and made sure each router advertised the networks of the routers around it. As a result, I had full end-to-end connectivity in my topology and was able to successfully conduct a traceroute command from the first router to the last multi-layer switch.

Below I have attached evidence of logs from my lab, which prove that my topology had multi-area OSPF functionality.

### **Lab Report Evidence:**

#### **R1#show ip route ospf**

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
level-2
        ia - IS-IS inter area, * - candidate default, U - per-user
static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l -
LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from
PfR
```

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks

O IA 10.0.2.0/30 [110/2] via 10.0.1.2, 01:39:40,  
GigabitEthernet0/0/0

O IA 10.0.3.0/30 [110/3] via 10.0.1.2, 01:29:40,  
GigabitEthernet0/0/0

O IA 10.0.4.0/30 [110/4] via 10.0.1.2, 01:27:51,  
GigabitEthernet0/0/0

O IA 10.0.5.0/30 [110/5] via 10.0.1.2, 01:23:21,  
GigabitEthernet0/0/0

O IA 10.0.6.0/30 [110/6] via 10.0.1.2, 01:09:39,  
GigabitEthernet0/0/0

**!!! The O IA indicated OSPF Inter-Area (OIA), which shows each route that is in a different area than R1.**

#### R5#show ipv6 route ospf

IPv6 Routing Table - default - 10 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2

IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP  
external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr -  
Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF  
ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application

O 2001:BD8:2::/64 [110/3]  
via FE80::CE7F:76FF:FE6A:A981, GigabitEthernet0/0/1

OI 2001:DB8:1::/64 [110/4]  
via FE80::CE7F:76FF:FE6A:A981, GigabitEthernet0/0/1

O 2001:DB8:2::/64 [110/3]  
via FE80::CE7F:76FF:FE6A:A981, GigabitEthernet0/0/1

O 2001:DB8:3::/64 [110/2]  
via FE80::CE7F:76FF:FE6A:A981, GigabitEthernet0/0/1

O 2001:DB8:6::/64 [110/2]  
via FE80::6E13:D5FF:FE20:BFD6, GigabitEthernet0/0/0

**!!! Similarly, OI is shown here as well, with the OI correlating to OSPF Interarea, as seen in the key.**

#### R5#show ip ospf database

OSPF Router with ID (5.5.5.5) (Process ID 1)

Router Link States (Area 0)

| Link ID<br>count | ADV Router | Age  | Seq#       | Checksum Link |
|------------------|------------|------|------------|---------------|
| 2.2.2.2          | 2.2.2.2    | 299  | 0x8000000C | 0x0043B8 1    |
| 3.3.3.3          | 3.3.3.3    | 1402 | 0x8000000E | 0x00DAE9 2    |
| 4.4.4.4          | 4.4.4.4    | 1436 | 0x80000009 | 0x00D2EB 2    |

|         |         |     |            |          |   |
|---------|---------|-----|------------|----------|---|
| 5.5.5.5 | 5.5.5.5 | 817 | 0x8000000C | 0x00AE02 | 2 |
| 6.6.6.6 | 6.6.6.6 | 469 | 0x80000007 | 0x006278 | 1 |

#### Net Link States (Area 0)

| Link ID  | ADV Router | Age  | Seq#       | Checksum |
|----------|------------|------|------------|----------|
| 10.0.2.2 | 3.3.3.3    | 144  | 0x80000004 | 0x004BC0 |
| 10.0.3.2 | 4.4.4.4    | 1436 | 0x80000003 | 0x007887 |
| 10.0.4.1 | 4.4.4.4    | 1436 | 0x80000003 | 0x00DB1C |
| 10.0.5.1 | 5.5.5.5    | 817  | 0x80000003 | 0x0007E3 |

#### Summary Net Link States (Area 0)

| Link ID | ADV Router | Age | Seq# | Checksum |
|---------|------------|-----|------|----------|
|---------|------------|-----|------|----------|

R5#show ipv6 ospf database

#### OSPFv3 Router with ID (5.5.5.5) (Process ID 10)

#### Router Link States (Area 0)

| ADV Router | Age  | Seq#       | Fragment ID | Link count | Bits |
|------------|------|------------|-------------|------------|------|
| 2.2.2.2    | 1905 | 0x8000000D | 0           | 1          | B    |
| 3.3.3.3    | 1420 | 0x80000011 | 0           | 2          | None |
| 4.4.4.4    | 1250 | 0x8000000D | 0           | 2          | None |
| 5.5.5.5    | 1049 | 0x8000000B | 0           | 1          | B    |

#### Net Link States (Area 0)

| ADV Router | Age  | Seq#       | Link ID | Rtr count |
|------------|------|------------|---------|-----------|
| 3.3.3.3    | 167  | 0x80000004 | 7       | 2         |
| 4.4.4.4    | 1495 | 0x80000003 | 6       | 2         |
| 4.4.4.4    | 1250 | 0x80000005 | 7       | 2         |

#### Inter Area Prefix Link States (Area 0)

| ADV Router | Age  | Seq#       | Prefix          |
|------------|------|------------|-----------------|
| 2.2.2.2    | 1905 | 0x80000003 | 2001:DB8:1::/64 |
| 5.5.5.5    | 1049 | 0x80000003 | 2001:DB8:5::/64 |
| 5.5.5.5    | 305  | 0x80000003 | 2001:DB8:6::/64 |

#### Link (Type-8) Link States (Area 0)

| ADV Router | Age  | Seq#       | Link ID | Interface |
|------------|------|------------|---------|-----------|
| 4.4.4.4    | 1495 | 0x80000004 | 7       | Gi0/0/1   |
| 5.5.5.5    | 1306 | 0x80000003 | 7       | Gi0/0/1   |

#### Intra Area Prefix Link States (Area 0)

| ADV Router<br>LSID | Age | Seq#       | Link ID | Ref-lsttype | Ref- |
|--------------------|-----|------------|---------|-------------|------|
| 3.3.3.3            | 167 | 0x80000004 | 7168    | 0x2002      | 7    |

|         |      |            |      |        |   |
|---------|------|------------|------|--------|---|
| 4.4.4.4 | 1495 | 0x80000003 | 6144 | 0x2002 | 6 |
| 4.4.4.4 | 1250 | 0x80000005 | 7168 | 0x2002 | 7 |

#### Router Link States (Area 2)

| ADV Router | Age  | Seq#       | Fragment ID | Link count | Bits |
|------------|------|------------|-------------|------------|------|
| 5.5.5.5    | 1049 | 0x8000000B | 0           | 1          | B    |
| 6.6.6.6    | 22   | 0x8000000E | 0           | 2          | None |
| 7.7.7.7    | 91   | 0x80000005 | 0           | 1          | None |

#### Net Link States (Area 2)

| ADV Router | Age  | Seq#       | Link ID | Rtr count |
|------------|------|------------|---------|-----------|
| 5.5.5.5    | 1049 | 0x80000004 | 6       | 2         |
| 6.6.6.6    | 22   | 0x80000003 | 38      | 2         |

#### Inter Area Prefix Link States (Area 2)

| ADV Router | Age  | Seq#       | Prefix          |
|------------|------|------------|-----------------|
| 5.5.5.5    | 1049 | 0x80000004 | 2001:DB8:4::/64 |
| 5.5.5.5    | 1049 | 0x80000004 | 2001:DB8:3::/64 |
| 5.5.5.5    | 1049 | 0x80000004 | 2001:DB8:2::/64 |
| 5.5.5.5    | 1049 | 0x80000004 | 2001:BD8:2::/64 |
| 5.5.5.5    | 1049 | 0x80000004 | 2001:DB8:1::/64 |

#### Link (Type-8) Link States (Area 2)

| ADV Router | Age  | Seq#       | Link ID | Interface |
|------------|------|------------|---------|-----------|
| 5.5.5.5    | 1049 | 0x80000007 | 6       | Gi0/0/0   |
| 6.6.6.6    | 1037 | 0x80000003 | 37      | Gi0/0/0   |

#### Intra Area Prefix Link States (Area 2)

| ADV Router<br>LSID | Age  | Seq#       | Link ID | Ref-lsttype | Ref- |
|--------------------|------|------------|---------|-------------|------|
| 5.5.5.5            | 1049 | 0x80000003 | 6144    | 0x2002      | 6    |
| 6.6.6.6            | 22   | 0x80000003 | 38912   | 0           |      |

### R5#show ip ospf border-routers

OSPF Router with ID (5.5.5.5) (Process ID 1)

#### Base Topology (MTID 0)

#### Internal Router Routing Table

Codes: i - Intra-area route, I - Inter-area route

```
i 2.2.2.2 [3] via 10.0.4.1, GigabitEthernet0/0/1, ABR, Area 0, SPF 14
i 6.6.6.6 [1] via 10.0.5.2, GigabitEthernet0/0/0, ABR, Area 0, SPF 14
```

**!!! This command shows the routers which are the border routers between areas. For this lab it was Router 2 and Router 6.**

### **Configs:**

#### **!! R1:**

```
R1#show run
Building configuration...
Current configuration : 3727 bytes
! Last configuration change at 16:00:35 UTC Thu Sep 11 2025
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
platform hardware throughput level 100000
hostname R1
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
!
address-family ipv6
exit-address-family
!
!
no aaa new-model
!
!
!
!
login on-success log
!
!
!
!
!
!
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
!
!
!
crypto pki trustpoint TP-self-signed-2663517976
enrollment selfsigned
```

subject-name cn=IOS-Self-Signed-Certificate-2663517976  
revocation-check none  
rsakeypair TP-self-signed-2663517976  
crypto pki certificate chain TP-self-signed-2663517976  
certificate self-signed 01  
30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101  
05050030  
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D  
43657274  
69666963 6174652D 32363633 35313739 3736301E 170D3235 30393131  
31353238  
35335A17 0D333030 31303130 30303030 305A3031 312F302D 06035504  
03132649  
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32  
36363335  
31373937 36308201 22300D06 092A8648 86F70D01 01010500 0382010F  
00308201  
0A028201 0100DA90 25588ED9 2C00C346 3FACAA5B 270CE011 E135ACBC  
5C404C79  
D84F4A75 F106B4CD 53F68FF4 E9FAB61C 55DD0A83 EA3E87B8 367E105F  
EA3A122F  
E30E948A A61AFB31 5AE21358 C2224602 19F35055 573FD738 5859A5EA  
DE3ED512  
19C435DF D2D6D9B4 76E4EDC7 D4998A36 FEE2457F 4746A7DE C33FA32C  
21B20320  
B5B1B072 1C2A7A17 B33B94EF 22D86ED2 57D2FBE8 224F0663 EC2AF0FF  
1B1F5E7A  
3B1000AD 67FB3782 0F91FBB0 2C8E0A5E BCFE662F 3B44F336 313B1E95  
AE1B29B5  
8EECCBC6 6C25AF9D CC6DF277 F1FDEBB1 A44C3E3C B0AE519A 6DA496A9  
E3A0898C  
7565A418 73EC487C A4750768 699A020B 2638E706 A1379E70 89E8104D  
1017DAD4  
0F02C520 382B0203 010001A3 53305130 0F060355 1D130101 FF040530  
030101FF  
301F0603 551D2304 18301680 14436482 D1A68A4E DBD389E2 879618F0  
96901241  
83301D06 03551D0E 04160414 436482D1 A68A4EDB D389E287 9618F096  
90124183  
300D0609 2A864886 F70D0101 05050003 82010100 13CAC16C 36B50085  
25881EF8  
5A78E684 E1D0C780 A35713F3 9241CAAE 824D810D C4FAAABC 36CE52D7  
282223CF  
8F46E6C8 EA7DF1C4 6B3EB9DE C77FBCCA 511B4C0A 3792B762 97A4BBAC  
D25D1353  
D66EAA71 78AE2BC3 DEEB085B 57B60455 A95C2D2E A6771554 AEFD5DF9  
0B1741B6  
7E064990 2F48BAEB F9F9B3B3 A0A0D175 32890419 1A9980FB 610A5E4D  
92A1C034  
E8128241 9C60A956 A0B9D533 C9006E28 86683E4C F75059BD 0F130B53  
1D82E67D



```
DB7ABD90 C0F93704 28158B57 BA2E38A5 1E3E0A90 63F1FF08 57EA68B6
00EB7948
6587DC28 08955F48 23DAD2A5 721FF725 EDCF5B8D B7E9F2FF A8B05DB2
CA27F91D
AC674404 F6E16574 51AA1C0E D01041B2 05381134
quit
!
license udi pid ISR4321/K9 sn FLM240607Q6
no license smart enable
diagnostic bootup level minimal
!
spanning-tree extend system-id
!
!
!
!
!
redundancy
mode none
!
!
!
!
!
!
!
!
interface GigabitEthernet0/0/0
ip address 10.0.1.1 255.255.255.252
ip ospf 1 area 1
negotiation auto
ipv6 address 2001:DB8:1::1/64
ipv6 ospf 10 area 1
!
interface GigabitEthernet0/0/1
no ip address
shutdown
negotiation auto
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
!
router ospf 1
router-id 1.1.1.1
network 10.0.1.0 0.0.0.3 area 1
!
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
```

```
ip tftp source-interface GigabitEthernet0
!
!
ipv6 router ospf 10
router-id 1.1.1.1
!
!
!
!
control-plane
!
!
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
!
!
!
!
!
!
end
R1#
```

## **R2:**

```
R2#show run
Building configuration...
Current configuration : 3930 bytes
!
! Last configuration change at 15:58:17 UTC Thu Sep 11 2025
!
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
!
hostname R2
!
boot-start-marker
boot-end-marker
!
!
vrf definition Mgmt-intf
!
address-family ipv4
exit-address-family
```

```
!  
address-family ipv6  
exit-address-family  
!  
!  
no aaa new-model  
!  
!  
!  
!  
login on-success log  
!  
!  
!  
!  
!  
!  
subscriber templating  
vtp domain cisco  
vtp mode transparent  
ipv6 unicast-routing  
multilink bundle-name authenticated  
!  
!  
!  
crypto pki trustpoint TP-self-signed-214699181  
enrollment selfsigned  
subject-name cn=IOS-Self-Signed-Certificate-214699181  
revocation-check none  
rsakeypair TP-self-signed-214699181  
!  
!  
crypto pki certificate chain TP-self-signed-214699181  
certificate self-signed 01  
3082032E 30820216 A0030201 02020101 300D0609 2A864886 F70D0101  
05050030  
30312E30 2C060355 04031325 494F532D 53656C66 2D536967 6E65642D  
43657274  
69666963 6174652D 32313436 39393138 31301E17 0D323530 39313131  
35323930  
385A170D 33303031 30313030 30303030 5A303031 2E302C06 03550403  
1325494F  
532D5365 6C662D53 69676E65 642D4365 72746966 69636174 652D3231  
34363939  
31383130 82012230 0D06092A 864886F7 0D010101 05000382 010F0030  
82010A02  
82010100 B19240E7 0C6E31FD 32492444 E84B5C61 7AD7597A 493364DE  
67CD3B8A  
B1F6ED84 DD012899 C1DF0056 EC990269 8252A7B3 D61CED22 07419CAE  
5776DB66
```

39E6A06E 944A9AEE 79AD3CA2 40E1634B 00095DB8 E1876E4A 8503F85C  
EEB93A94  
CA23BAD5 202F7485 473210D1 84A8A8EE 3533B1AA 716AB9C5 ACA8DFB4  
1EF06233  
65CFA246 2BAD38EF E7B6B61B 8675A87B 97F4BE7E 9165F5B7 10688E14  
1911CAD9  
31D2C657 97265BC4 3C7F980A BCD82DCB AAAEE5A9 080D1377 727B13ED  
01E9E320  
8EC125C6 F9A02871 302AA2A8 F4CA3C60 2F1A49EA 72C88473 0B138630  
159C9DBD  
2786F1F3 8678AAFF 7C88A8D9 A5C919BF AA04AE26 878B00C1 3B0B1830  
DB017F40  
3B816225 02030100 01A35330 51300F06 03551D13 0101FF04 05300301  
01FF301F  
0603551D 23041830 1680148A 93C625EC 4D1D6FC2 33AD0A76 8640A35C  
FDA95D30  
1D060355 1D0E0416 04148A93 C625EC4D 1D6FC233 AD0A7686 40A35CFD  
A95D300D  
06092A86 4886F70D 01010505 00038201 010002BF 834F2CE9 89CF1656  
C0BA05EF  
54326B14 9FC9E064 02C38897 4DCCCD21 A1BF4DF8 5189FF80 0AC33ECB  
941D6FB8  
329B31AF 5C07DF88 148B513C 866AD26A 1F2323E2 67A276AD A2D0BD06  
8B35939E  
B38C0083 B1CEBCD7 80105D80 87BFC6E5 4216F32A A642CEC7 FB40B4BB  
57A31B73  
941EB7B8 B0E7D619 930B0322 E67F6D54 872E94A8 06680163 3A6509CD  
5B5534F0  
0042E864 D30ABBFD 8DE6A1AC 8B7A297B D3B77DA6 D303747B 56602359  
0C75194B  
04D7A472 882E2377 6085B752 3E08614D BE2E076A 19C7C88B B41B3370  
8C8D109F  
40DE5A46 CD059974 E93EBA64 98F08254 B4409E10 D183B29F EB570966  
FB082206  
9F52C4D0 FD5B77AB A966A1B3 4C64D14F D32F  
quit  
!  
license udi pid ISR4321/K9 sn FDO21442167  
no license smart enable  
diagnostic bootup level minimal  
!  
spanning-tree extend system-id  
!  
!  
!  
!  
redundancy  
mode none  
!  
!  
!  
!

```
!  
!  
!  
!  
interface GigabitEthernet0/0/0  
ip address 10.0.1.2 255.255.255.252  
ip ospf 1 area 1  
negotiation auto  
ipv6 address 2001:DB8:1::2/64  
ipv6 ospf 10 area 1  
!  
interface GigabitEthernet0/0/1  
ip address 10.0.2.1 255.255.255.252  
ip ospf 1 area 0  
negotiation auto  
ipv6 address 2001:DB8:2::1/64  
ipv6 ospf 10 area 0  
!  
interface Serial0/1/0  
no ip address  
shutdown  
!  
interface Serial0/1/1  
no ip address  
shutdown  
!  
interface GigabitEthernet0  
vrf forwarding Mgmt-intf  
no ip address  
shutdown  
negotiation auto  
!  
router ospf 1  
router-id 2.2.2.2  
network 10.0.1.0 0.0.0.3 area 1  
network 10.0.2.0 0.0.0.3 area 0  
!  
ip forward-protocol nd  
ip http server  
ip http authentication local  
ip http secure-server  
ip tftp source-interface GigabitEthernet0  
!  
!  
ipv6 router ospf 10  
router-id 2.2.2.2  
!  
ipv6 router ospf 1  
router-id 2.2.2.2  
!  
!  
!
```

```
!  
!  
control-plane  
!  
!  
line con 0  
transport input none  
stopbits 1  
line aux 0  
stopbits 1  
line vty 0 4  
login  
!  
!  
!  
!  
!  
!  
end  
!!! R3:  
R3#sho run  
Building configuration...  
Current configuration : 3922 bytes  
!  
! Last configuration change at 15:43:47 UTC Thu Sep 11 2025  
!  
version 16.9  
service timestamps debug datetime msec  
service timestamps log datetime msec  
platform qfp utilization monitor load 80  
platform punt-keepalive disable-kernel-core  
!  
hostname R3  
!  
boot-start-marker  
boot-end-marker  
!  
!  
vrf definition Mgmt-intf  
!  
address-family ipv4  
exit-address-family  
!  
address-family ipv6  
exit-address-family  
!  
!  
no aaa new-model  
!  
!  
!
```

```
!  
!  
!  
!  
!  
!  
!  
login on-success log  
!  
!  
!  
!  
!  
!  
subscriber templating  
!  
!  
!  
!  
vtp domain cisco  
vtp mode transparent  
ipv6 unicast-routing  
multilink bundle-name authenticated  
!  
!  
!  
!  
crypto pki trustpoint TP-self-signed-2949602955  
enrollment selfsigned  
subject-name cn=IOS-Self-Signed-Certificate-2949602955  
revocation-check none  
rsa-keypair TP-self-signed-2949602955  
!  
!  
crypto pki certificate chain TP-self-signed-2949602955  
certificate self-signed 01  
30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101  
05050030  
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D  
43657274  
69666963 6174652D 32393439 36303239 3535301E 170D3235 30393131  
31353136  
35375A17 0D333030 31303130 30303030 305A3031 312F302D 06035504  
03132649  
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32  
39343936  
30323935 35308201 22300D06 092A8648 86F70D01 01010500 0382010F  
00308201  
0A028201 01009951 ABD8446A 030D923F D0E7657F AB738A23 52CF8F11  
E7E3AEC4
```

62607555 349C460D 924F3224 334C0328 C91511F0 7CC94047 0FD74279  
89486B6F  
C2379F4D BD7562FB 6FA77437 74F2C5D0 AF192184 3FE7F0A0 782B599F  
D0774B55  
20DCDF57 6BBF0A36 4104C3CD B3743674 382D8FE9 5D27DE7C F661BED0  
3BFDCE16  
587F0030 F3803113 5FC7F1EB CB423DDE 33D2095A 7CF17CB0 473BA25A  
DA3794DD  
50FB997F 31A1A44A 392EE7D3 D0C62D26 0D246D2D 30B3DAB3 2320F3F9  
A670D1DE  
A140EB41 EAE29FAD 839C78C6 A34D966D 99B29F09 BB076557 81B873CB  
FBB96D65  
7A801003 15BEB80E A6880C94 252E6FA0 8FD04EF1 53897C18 B84360E5  
0BC9261D  
9FB77C36 769F0203 010001A3 53305130 0F060355 1D130101 FF040530  
030101FF  
301F0603 551D2304 18301680 149F889E 5CDB3901 5F1370D1 6AA517E0  
86C9F7C5  
B6301D06 03551D0E 04160414 9F889E5C DB39015F 1370D16A A517E086  
C9F7C5B6  
300D0609 2A864886 F70D0101 05050003 82010100 28E285CA 11A68950  
C527989B  
146D46E9 731DCDA9 20199E46 A84D8B55 E47CD92C FDBF1FDA C2BA3830  
E4B0B747  
524219F3 A799DF36 E0284DEE F0F887E7 D80D9940 71EDD9AB 6C2C9B8D  
1B428371  
69644EA0 ACA74D0F E578285A A0CC6BD4 51C73531 E06B7961 8BAA1EEB  
67EFBB46  
7F3ED1B0 5821DDB6 3F671489 8AE591B0 5F2815F5 0F5796F2 257BB238  
373D99A8  
4E6530DD 2A959E79 720278C0 0021030E 7E459084 716AABAE A5CD4B86  
94DDEFD0  
B9C5B028 91EBC281 BC703D8D 6FCA6D46 70F23A98 F14D0BE7 66F990EB  
B38197B2  
A38D8CA9 FD5AEE26 641885C0 E901D6B3 7746828F ED330F40 38EAE529  
19F96D4C  
55F4D93C 40B3E04B 3F801741 4E7D0594 B07D1663

quit

!

license udi pid ISR4321/K9 sn FDO214420HW

license boot level appxk9

no license smart enable

diagnostic bootup level minimal

!

spanning-tree extend system-id

!

!

!

!

redundancy

mode none

!



```
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0/0  
ip address 10.0.3.1 255.255.255.252  
negotiation auto  
ipv6 address 2001:DB8:3::1/64  
ipv6 ospf 10 area 0  
!  
interface GigabitEthernet0/0/1  
ip address 10.0.2.2 255.255.255.252  
negotiation auto  
ipv6 address 2001:BD8:2::2/64  
ipv6 ospf 10 area 0  
!  
interface Serial0/1/0  
no ip address  
shutdown  
!  
interface Serial0/1/1  
no ip address  
shutdown  
!  
interface GigabitEthernet0  
vrf forwarding Mgmt-intf  
no ip address  
shutdown  
negotiation auto  
!  
router ospf 1  
router-id 3.3.3.3  
network 10.0.2.0 0.0.0.3 area 0  
network 10.0.3.0 0.0.0.3 area 0  
!  
ip forward-protocol nd  
ip http server  
ip http authentication local  
ip http secure-server  
ip tftp source-interface GigabitEthernet0  
!  
!  
ipv6 router ospf 10  
router-id 3.3.3.3  
!  
!
```

```
!  
!  
!  
control-plane  
!  
!  
line con 0  
transport input none  
stopbits 1  
line aux 0  
stopbits 1  
line vty 0 4  
login
```

```
!  
!  
!  
!  
!  
!  
end
```

```
R3#
```

```
!!! R4:
```

```
R4#show run  
Building configuration...  
Current configuration : 3916 bytes  
!  
! Last configuration change at 15:55:07 UTC Thu Sep 11 2025  
!  
version 16.9  
service timestamps debug datetime msec  
service timestamps log datetime msec  
platform qfp utilization monitor load 80  
platform punt-keepalive disable-kernel-core
```

```
!  
hostname R4  
!  
boot-start-marker  
boot-end-marker  
!  
!  
vrf definition Mgmt-intf  
!  
address-family ipv4  
exit-address-family  
!  
address-family ipv6  
exit-address-family  
!  
!  
no aaa new-model  
!
```

```
!  
!  
!  
login on-success log  
!  
!  
!  
!  
!  
!  
subscriber templating  
vtp domain cisco  
vtp mode transparent  
ipv6 unicast-routing  
multilink bundle-name authenticated  
!  
!  
!  
crypto pki trustpoint TP-self-signed-2667303412  
enrollment selfsigned  
subject-name cn=IOS-Self-Signed-Certificate-2667303412  
revocation-check none  
rsakeypair TP-self-signed-2667303412  
!  
!  
crypto pki certificate chain TP-self-signed-2667303412  
certificate self-signed 01  
30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101  
05050030  
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D  
43657274  
69666963 6174652D 32363637 33303334 3132301E 170D3235 30393131  
31353337  
33355A17 0D333030 31303130 30303030 305A3031 312F302D 06035504  
03132649  
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32  
36363733  
30333431 32308201 22300D06 092A8648 86F70D01 01010500 0382010F  
00308201  
0A028201 01009865 507D993F FCC3AFD0 E61A61DD 96A40B8F A240DC14  
6B736760  
60F7BC18 0D1A61B7 179362CF E8DB7FEA 7ACEB3D3 002AE791 A9727CB7  
A08A5F40  
9A612CF1 DCBF7D7F B249B453 4EA1A3AE 2DC2785F 26BF4F04 12AB8A15  
72F4499D  
FC3E9520 D063409E 6372A6F0 AE69A128 70942447 EE89A0E8 AB30F1A1  
59FF4AF4  
51658BE1 911538C3 EB89DC4E 95CB1D4F 52A7EF7D 0960530E A8014E3B  
29A3E22B  
713807B5 274E5DAA 834E4E9B 1935139F 9FDB5599 07B7A2FE 1C904B8F  
F1DE5E7F
```

```
C2A2F965 07969042 B9BBB492 C833EFAC 7D0FE63B 4B250E8C 126ACDE1
54E26B5C
381C3D97 222E9268 7B2FBC24 EDF86CBA 2202AEE0 57784F25 DE683D37
966FDAE6
2B31FBDD AF470203 010001A3 53305130 0F060355 1D130101 FF040530
030101FF
301F0603 551D2304 18301680 14369687 F63E1630 74D559F2 475F5826
7D7D33DA
81301D06 03551D0E 04160414 369687F6 3E163074 D559F247 5F58267D
7D33DA81
300D0609 2A864886 F70D0101 05050003 82010100 619837D5 D3EB6DF6
5D4C2FDA
0A05A47B 51EC93E0 0F8BF7D1 368F485D 57D1D0D8 3FD11AE4 553EAF48
D037ED9F
89E53680 3174BAB1 5672F175 2CF23000 F8EBFFD9 772B4D50 300268A3
C630E6A8
42E986D0 CD3790C1 05A697E7 F74C59B4 8FF9E748 0652EFC1 C25103CF
50F46E83
E6E24FE7 446A8D9A 48BE6C9B C5022271 8D9863C6 F41931A8 38789413
88DA13A5
BD9E0A2B F4458DB1 2051B0C8 2FA78B34 495C5208 51D638E4 CD463FDC
B9120929
420B8DAC 00647C9E BECFB192 008C74B2 861F467A 9A322789 F3B1EE9D
BB6E76D2
62D4CC44 A708588C 3D268501 A929A221 1F7A405B 81469163 2A725654
C994BA38
07C3C35B 29380845 F5D295EC 598E1905 4AC9F869
quit
!
license udi pid ISR4321/K9 sn FLM2407011F
no license smart enable
diagnostic bootup level minimal
!
spanning-tree extend system-id
!
!
!
!
!
redundancy
mode none
!
!
!
!
!
!
!
!
!
interface GigabitEthernet0/0/0
ip address 10.0.3.2 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:3::2/64
```

```
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/0/1
ip address 10.0.4.1 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:4::1/64
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/1/0
no ip address
shutdown
negotiation auto
!
interface GigabitEthernet0/1/1
no ip address
shutdown
negotiation auto
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
!
router ospf 1
router-id 4.4.4.4
network 10.0.3.0 0.0.0.3 area 0
network 10.0.4.0 0.0.0.3 area 0
!
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
!
!
ipv6 router ospf 10
router-id 4.4.4.4
!
!
!
!
!
control-plane
!
!
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
```

```
login
!
!
!
!
!
!
!
end
!!! R5:
R5#show run
Building configuration...
Current configuration : 1807 bytes
!
! Last configuration change at 16:21:05 UTC Thu Sep 11 2025
!
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
no platform punt-keepalive disable-kernel-core
!
hostname R5
!
boot-start-marker
boot system bootflash:isr4300-universalk9.16.09.08.SPA.bin
boot system flash isr4300-universalk9.16.09.08.SPA.bin
boot-end-marker
!
!
vrf definition Mgmt-intf
!
address-family ipv4
exit-address-family
!
address-family ipv6
exit-address-family
!
!
no aaa new-model
!
!
!
!
!
login on-success log
!
!
!
!
!
```

```
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
!
!
!
!
!
license udi pid ISR4321/K9 sn FLM240607Q1
no license smart enable
diagnostic bootup level minimal
!
spanning-tree extend system-id
!
!
!
!
!
redundancy
mode none
!
!
!
!
!
!
!
!
!
interface GigabitEthernet0/0/0
ip address 10.0.5.1 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:5::1/64
ipv6 ospf 10 area 2
!
interface GigabitEthernet0/0/1
ip address 10.0.4.2 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:4::2/64
ipv6 ospf 10 area 0
!
interface GigabitEthernet0/1/0
no ip address
shutdown
negotiation auto
!
interface GigabitEthernet0/1/1
no ip address
shutdown
negotiation auto
!
interface GigabitEthernet0
```

```
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
!
router ospf 1
router-id 5.5.5.5
network 10.0.4.0 0.0.0.3 area 0
network 10.0.5.0 0.0.0.3 area 0
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
!
ipv6 router ospf 10
router-id 5.5.5.5
!
!
!
!
!
control-plane
!
!
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
!
!
!
!
!
end
```

### !!! MLS1:

```
SW1#show run
Building configuration...
Current configuration : 8757 bytes
!
! Last configuration change at 15:36:48 UTC Thu Sep 11 2025
!
version 16.9
no service pad
service timestamps debug datetime msec
```



```
service timestamps log datetime msec
! Call-home is enabled by Smart-Licensing.
service call-home
no platform punt-keepalive disable-kernel-core
!
hostname SW1
!
!
vrf definition Mgmt-vrf
!
address-family ipv4
exit-address-family
!
address-family ipv6
exit-address-family
!
!
!
!
!
no aaa new-model
switch 1 provision c9200l-24t-4g
!
!
!
!
!
!
!
!
!
ip routing
ip arp entry learn 10240
!
!
!
!
login on-success log
ipv6 unicast-routing
call-home
! If contact email address in call-home is configured as sch-smart-
licensing@cisco.com
! the email address configured in Cisco Smart License Portal will be
used as contact email address to send SCH notifications.
contact-email-addr sch-smart-licensing@cisco.com
profile "CiscoTAC-1"
active
destination transport-method http
no destination transport-method email
!
crypto pki trustpoint SLA-TrustPoint
enrollment pkcs12
```

```
revocation-check crl
!
crypto pki trustpoint TP-self-signed-226317576
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-226317576
revocation-check none
rsakeypair TP-self-signed-226317576
!
!
crypto pki certificate chain SLA-TrustPoint
certificate ca 01
30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101
0B050030
32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317
43697363
6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533
30313934
3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355
040A1305
43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73
696E6720
526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382
010F0030
82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1
F1EFF64D
CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388
8A38E520
1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBEA3 700A8BF7
D8F256EE
4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F
EA2956AC
7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C0BD23CF
58BD7188
68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B
42C68BB7
C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8
8F27D191
C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368
95135E44
DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04
04030201
06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604
1449DC85
4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01
010B0500
03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78
240DA905
604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1
6C9E3D8B
D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146
8DFC66A8
```

467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2  
55A9232C  
7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49  
1765308B  
5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69  
39F08678  
80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD  
230E3AFB  
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F  
719BB2F0  
D697DF7F 28  
quit  
crypto pki certificate chain TP-self-signed-226317576  
certificate self-signed 01  
3082032E 30820216 A0030201 02020101 300D0609 2A864886 F70D0101  
05050030  
30312E30 2C060355 04031325 494F532D 53656C66 2D536967 6E65642D  
43657274  
69666963 6174652D 32323633 31373537 36301E17 0D323530 39313131  
35323031  
335A170D 33303031 30313030 30303030 5A303031 2E302C06 03550403  
1325494F  
532D5365 6C662D53 69676E65 642D4365 72746966 69636174 652D3232  
36333137  
35373630 82012230 0D06092A 864886F7 0D010101 05000382 010F0030  
82010A02  
82010100 A36D097E 6B8491A0 AD94A6DB CF62981C F260C92F 313D01B0  
A9634334  
3429DB04 A13CFE2A 41CD2D5B 77C90503 3642A1C6 F3DB96CA 70A327D7  
4680D22F  
81B791E0 FFA2858A 14612D4B 55A24B1C 7026FA71 4E08A3DF 20D74A05  
1A3ADE91  
856B21E2 5C776BB2 2EADDB84 44C88E2A 0A6DB961 8D7F9167 E5179CCC  
3DB86DAD  
8D7C65E3 3315A664 3056A61B A5A08BD1 57AF412E 228AA209 50DCA82B  
3855AB0E  
72A558B8 357437DB A2556E11 2C9C40FE A3032BE1 4BB63FCD A13E5951  
792BFF26  
6A9E59BF 9D8251F3 783774B6 94437806 CC86CB74 50A00421 D17938E5  
B201504E  
EF0C40C2 0FA556D7 B80267F5 FC61780E 503834FC 5BCB22DF EE6F181D  
7C3319DB  
7CADAD7F 02030100 01A35330 51300F06 03551D13 0101FF04 05300301  
01FF301F  
0603551D 23041830 16801439 43637D82 7D9267D7 00F62055 2A706A09  
F95A1930  
1D060355 1D0E0416 04143943 637D827D 9267D700 F620552A 706A09F9  
5A19300D  
06092A86 4886F70D 01010505 00038201 01004410 C7FC7EC8 E6CE307B  
39781B1C  
ECBB0E07 9EB99BEE 9DDE3AE4 F6D6F570 AA77252A E7C11E43 BE5BDAB8  
34B2DAC8

```
A5897E26 08875D68 B9678CD9 0719CC7D 12E48479 72ABD6FB 7B5FA63F
7187B8D4
409DEF76 15277BDA BD1380CC 3ABCFABA 86624A3C 5E9B0BCA E53BA77D
74F73045
B389593E 578F582C 68219BDA B9C8D60E A7E29B89 050D198B 383F126B
23A8210C
E27515F6 92D576D5 FAEDE51B DDD1CD0D 7FDDDB17 75FB9A37 AD5394DD
FC21E171
866643F7 E5120388 1E19A1A8 BE126430 BE6810A3 D54113C1 0B547206
5ABE67A8
51AD19C1 867271FE 4AB250B6 F2F214DA 909F98C7 CC958E6C 98B40CFC
1807CE25
EC0AA00B 9D26A56A 16C3E3AE 1AE74E4B 4362
quit
!
license boot level network-essentials addon dna-essentials
!
!
diagnostic bootup level minimal
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
!
!
redundancy
mode sso
!
!
!
class-map match-any system-cpp-police-topology-control
description Topology control
class-map match-any system-cpp-police-sw-forward
description Sw forwarding, L2 LVX data, LOGGING
class-map match-any system-cpp-default
description Inter FED, EWLC control, EWLC data
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description Learning cache ovfl, High Rate App, Exception, EGR
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class-map match-any system-cpp-police-forus
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class-map match-any system-cpp-police-multicast-end-station
description MCAST END STATION
class-map match-any system-cpp-police-multicast
description Transit Traffic and MCAST Data
class-map match-any system-cpp-police-l2-control
description L2 control
class-map match-any system-cpp-police-dot1x-auth
```

```
description DOT1X Auth
class-map match-any system-cpp-police-data
description ICMP redirect, ICMP_GEN and BROADCAST
class-map match-any system-cpp-police-stackwise-virt-control
description Stackwise Virtual
class-map match-any non-client-nrt-class
class-map match-any system-cpp-police-routing-control
description Routing control and Low Latency
class-map match-any system-cpp-police-protocol-snooping
description Protocol snooping
class-map match-any system-cpp-police-dhcp-snooping
description DHCP snooping
class-map match-any system-cpp-police-system-critical
description System Critical and Gold Pkt
!
policy-map system-cpp-policy
!
!
!
!
!
!
!
!
!
!
!
!
!
interface GigabitEthernet0/0
vrf forwarding Mgmt-vrf
no ip address
shutdown
negotiation auto
!
interface GigabitEthernet1/0/1
!
interface GigabitEthernet1/0/2
no switchport
ip address 10.0.5.2 255.255.255.252
ipv6 address 2001:DB8:5::2/64
ipv6 ospf 10 area 2
!
interface GigabitEthernet1/0/3
no switchport
ip address 10.0.6.1 255.255.255.252
ipv6 address 2001:DB8:6::1/64
ipv6 ospf 10 area 2
interface GigabitEthernet1/0/4
interface GigabitEthernet1/0/5
interface GigabitEthernet1/0/6
interface GigabitEthernet1/0/7
```

```
interface GigabitEthernet1/0/
interface GigabitEthernet1/0/9
interface GigabitEthernet1/0/10
interface GigabitEthernet1/0/11
interface GigabitEthernet1/0/1
interface GigabitEthernet1/0/13
interface GigabitEthernet1/0/14
interface GigabitEthernet1/0/15
interface GigabitEthernet1/0/16
interface GigabitEthernet1/0/17
interface GigabitEthernet1/0/18
interface GigabitEthernet1/0/19
interface GigabitEthernet1/0/20
interface GigabitEthernet1/0/21
interface GigabitEthernet1/0/2
interface GigabitEthernet1/0/23
interface GigabitEthernet1/0/24
interface GigabitEthernet1/1/1
interface GigabitEthernet1/1/2
interface GigabitEthernet1/1/3
interface GigabitEthernet1/1/4
interface Vlan1
no ip address
shutdown
router ospf 1
router-id 6.6.6.6
network 10.0.5.0 0.0.0.3 area 0
network 10.0.6.0 0.0.0.3 area 2
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ipv6 router ospf 10
router-id 6.6.6.6
control-plane
service-policy input system-cpp-policy
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
line vty 5 15
login
end
SW1#
```

## **SW2:**

```
S2#show run
Building configuration...
Current configuration : 8701 bytes
```

```
! Last configuration change at 15:43:48 UTC Thu Sep 11 2025
version 16.9
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
! Call-home is enabled by Smart-Licensing.
service call-home
no platform punt-keepalive disable-kernel-core
hostname S2
vrf definition Mgmt-vrf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
switch 1 provision c9200l-24t-4g
ip routing
ip arp entry learn 10240
login on-success log
ipv6 unicast-routing
call-home
! If contact email address in call-home is configured as sch-smart-
licensing@cisco.com
! the email address configured in Cisco Smart License Portal will be
used as contact email address to send SCH notifications.
contact-email-addr sch-smart-licensing@cisco.com
profile "CiscoTAC-1"
active
destination transport-method http
no destination transport-method email
crypto pki trustpoint TP-self-signed-11466316
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-11466316
revocation-check none
rsa-keypair TP-self-signed-11466316
crypto pki trustpoint SLA-TrustPoint
enrollment pkcs12
revocation-check crl
crypto pki certificate chain TP-self-signed-11466316
certificate self-signed 01
3082032C 30820214 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
2F312D30 2B060355 04031324 494F532D 53656C66 2D536967 6E65642D
43657274
69666963 6174652D 31313436 36333136 301E170D 32353039 31303137
31343438
5A170D33 30303130 31303030 3030305A 302F312D 302B0603 55040313
24494F53
2D53656C 662D5369 676E6564 2D436572 74696669 63617465 2D313134
36363331
36308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
0A028201
```

0100AEA9 87D92BF9 BCFE262B 2A3250D1 68965733 6D616D71 5151EB22  
22CEC5BC  
7AE2D915 F5C7344E 67CB8DCA 8512F12E 39DC0CA0 828BECEC ACDB10C9  
529E775F  
C9435728 A660D0C9 DD011245 37224162 0C7FDD9B C2E2DB39 0856455D  
DE08D031  
9D09843A 28531C9B E863A63E D31AF9DD FC64015B 1A64FFCC F573B013  
D0E82DCA  
6055ADAB 26B91849 07786B4D 4F78211B 6DF2F157 25114984 2B767295  
4C5C5400  
5194A5A0 A784A2BE B1CC57CB 5742830F E53DA1AF 7EA31A6D C47FE994  
FE98BB11  
CF5AA35D 8F6CF691 8EDED35D C6F0AF1D 74B10210 1B8956E4 BF96ABCD  
19A472CA  
FC6D93CB FEBF80C7 D0B63D06 96070A5F 7B2282B5 14AA1E72 4433926E  
0D99A3CE  
EC430203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF  
301F0603  
551D2304 18301680 1471CD1D 4A727376 FAE0CE09 38516A3C C42FE8E5  
63301D06  
03551D0E 04160414 71CD1D4A 727376FA E0CE0938 516A3CC4 2FE8E563  
300D0609  
2A864886 F70D0101 05050003 82010100 058A3BED 63001D9D 14A55AB8  
A5BF3046  
6724CF76 7A0FB42E BDEFD3A7 E895584C 87FFFD08 29A641E2 B111DB11  
56FF4D3E  
C3679AE5 B8709758 ADEE6491 0A9FD881 CDC2C365 D883337C 3157F994  
90BB15F1  
54BCBA37 90E559D3 EC908F13 26F07CB7 FA83AFEB BB39F587 2EF40582  
7D398D17  
98EEA507 03FA9BB2 4FF2252E 0446AE8C A713E08D B9C6666C DB9AC24C  
956886B7  
C76816D6 873D2A34 D3E8AC57 A2AF4D87 4170CC5B D4F0EA0B 0D048AE1  
9CE7CCB9  
7312FDEE CC4DC231 22DD3ADA EED6A828 69711B00 C922C3C8 8068C20E  
F6D7327E  
65327AA6 09DD18B4 6A9A2846 3CD0FDB3 BE947BA7 1173FC74 87A1DBC3  
03F583E7  
E2955421 1924D6CC 1BB8264A 6A3A7EC1  
quit  
crypto pki certificate chain SLA-TrustPoint  
certificate ca 01  
30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101  
0B050030  
32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317  
43697363  
6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533  
30313934  
3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355  
040A1305  
43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73  
696E6720



526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382  
010F0030  
82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1  
F1EFF64D  
CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388  
8A38E520  
1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBEA3 700A8BF7  
D8F256EE  
4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F  
EA2956AC  
7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C0BD23CF  
58BD7188  
68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B  
42C68BB7  
C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8  
8F27D191  
C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368  
95135E44  
DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04  
04030201  
06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604  
1449DC85  
4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01  
010B0500  
03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78  
240DA905  
604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1  
6C9E3D8B  
D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146  
8DFC66A8  
467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2  
55A9232C  
7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49  
1765308B  
5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69  
39F08678  
80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD  
230E3AFB  
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F  
719BB2F0  
D697DF7F 28

quit

license boot level network-essentials addon dna-essentials

diagnostic bootup level minimal

spanning-tree mode rapid-pvst

spanning-tree extend system-id

redundancy

mode sso

class-map match-any system-cpp-police-topology-control

description Topology control

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description Sw forwarding, L2 LVX data, LOGGING

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Exception, NFL SAMPLED DATA, RPF Failed
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class-map match-any system-cpp-police-system-critical
description System Critical and Gold Pkt
policy-map system-cpp-policy
interface GigabitEthernet0/0
vrf forwarding Mgmt-vrf
no ip address
shutdown
negotiation auto
interface GigabitEthernet1/0/1
interface GigabitEthernet1/0/2
interface GigabitEthernet1/0/3
no switchport
no ip address
interface GigabitEthernet1/0/4
no switchport
ip address 10.0.6.2 255.255.255.252
ipv6 address 2001:DB8:6::2/64
ipv6 ospf 10 area 2
interface Vlan1
ip address dhcp
router ospf 1
router-id 7.7.7.7
```

```

network 10.0.6.0 0.0.0.3 area 2
!
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip http client source-interface Vlan1
ip tftp source-interface Vlan1
ipv6 router ospf 10
router-id 7.7.7.7
control-plane
service-policy input system-cpp-policy
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
line vty 5 15
login
End

```

## Lab Commands:

### OSPFv2 (IPv4)

**router ospf** *process-id*

Example: *router ospf 1*

**Purpose:** Starts the OSPF process on the router with a locally defined process ID.

**network** *ip-address wildcard-mask area area-id*

Example: *network 10.0.5.0 0.0.0.3 area 0*

**Purpose:** Specifies which interfaces/networks participate in OSPF and assigns them to a particular OSPF area.

**ip ospf** *process-id area area-id*

Example: *ip ospf 1 area 0*

**Purpose:** Enables OSPF on a specific interface and assigns it to an OSPF area.

### ip routing

Example: *ip routing*

**Purpose:** Enables IPv4 packet forwarding globally; required for OSPF to function on the router.

**interface** *type slot/port*

Example: *interface GigabitEthernet1/0/2*

**Purpose:** Enters interface configuration mode to apply IP addresses and OSPF settings.

**ip address** *ip-address subnet-mask*

Example: *ip address 10.0.5.2 255.255.255.252*

**Purpose:** Assigns an IPv4 address and subnet mask to the interface so OSPF can advertise it.

### OSPFv3 (IPv6)

### **ipv6 unicast-routing**

Example: *ipv6 unicast-routing*

**Purpose:** Enables global IPv6 routing; required for OSPFv3 to run.

### **router ospfv3 process-id**

Example: *router ospfv3 1*

**Purpose:** Starts the OSPFv3 process on the router with a local process ID.

### **interface type slot/port**

Example: *interface GigabitEthernet1/0/3*

**Purpose:** Enters interface configuration mode to assign IPv6 addresses and enable OSPFv3.

### **ipv6 address ipv6-address/prefix-length**

Example: *ipv6 address 2001:db8:6::1/64*

**Purpose:** Assigns an IPv6 address to the interface so OSPFv3 can advertise it.

### **ipv6 enable**

Example: *ipv6 enable*

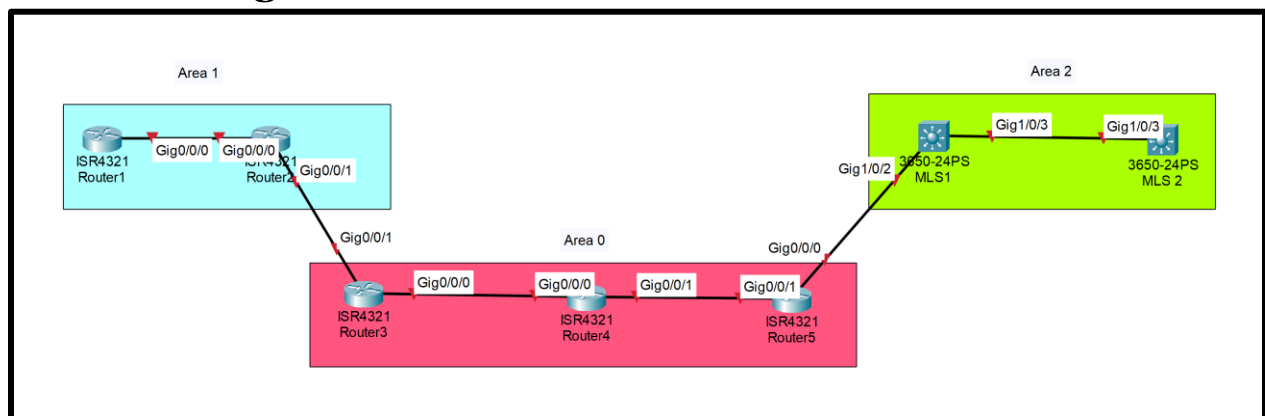
**Purpose:** Enables IPv6 processing on the interface (creates a link-local address if none exists).

### **ipv6 ospf process-id area area-id**

Example: *ipv6 ospf 1 area 0*

**Purpose:** Enables OSPFv3 on the interface and assigns it to a specific OSPF area.

## **Network Diagram:**



## **Problems:**

### **1. Broken Power Supply Cords**

After beginning the lab, I noticed one of my routers wouldn't show anything in PuTTY. After beginning with looking at the physical router, I noticed the light which indicated if the router was receiving power or not was off. We investigated the wiring of the routers and located the power supply brick and power supply cord. We noticed that the power

brick had its lights off because of no power. We wanted to narrow the problem, so we had switched the power cord of a different power supply brick with the broken one and noticed that the issue switched to the other brick, confirming the problem with the power supply cord. We then threw out the broken power supply cord and replaced it with a new one, which fixed the issue.

## **2. No IOS Boot Image**

This was certainly the most tedious issue. We noticed that one of the routers would not load its IOS Boot image and would repeatedly locate the iOS image but couldn't load it at some point within the loading process.

My first idea was to load an IOS image off a working router to a USB and copy the file from the USB to the Router's flash. We found out shortly after that the router couldn't locate its flash, and we then tried to boot from the USB itself, which had initially worked, but it came across the same issue where it located the IOS image file but couldn't load it.

The second idea was to download tftpd64, and set up my PC as a TFTP Server. I transferred my iOS boot image file into my TFTP Root Directory. I changed my PC's and Router's IP Address so they could immediately communicate with each other because the router did not use DHCP, and I disabled all firewalls to allow communication. We set our IP Address, TFTP Server, TFTP File, and BOOT variables then used the boot command to boot from the tftp file off the tftp server, but the same problem appeared where it could locate the iOS Image but couldn't run it.

Finally, after the help of Mr. Mason, we decided that because of the issues at hand, it was either a hardware issue with the router or an old version of ROMMON which couldn't support up-to-date commands needed to run a fresh iOS image such as the **tftpdnld** command.

## **3. No Power to MLS**

I noticed that for one of my MLS's, it wouldn't turn on or show anything when consoled into. After looking at the physical switch, it was a power problem since I didn't see any lights turned on. After some work and looking at the setup of the rack, I found out that there was a cable which wasn't connected into the router.

## **4. Console Cabling into MLS**

We also noticed that even when the MLS had power I still couldn't console into it. After looking at the same MLS version but unracked, I realized that the console cable was in

the back of the MLS in a hard-to-reach place. I looked at different working MLS's and found the console cable port.

**Conclusion:**

This was a very valuable and successful lab in the end. I learned a lot about OSPF, and refreshed and reviewed my current knowledge on single-area OSPF. This lab is very valuable because OSPF is one of the most popular and used routing protocols in the entire world, and it is extremely important for network administrators to practically understand and be able to configure not only the single-area function, but the multi-area function which is widely used in larger topologies.